

ClaudeOS — Client Documentation

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What Is ClaudeOS?

ClaudeOS is an AI Operating System — a secure, multi-user coordination layer that unifies AI agents, memory, workflows, and project management into a single authenticated control center. It runs on your local machine (Windows 10) and exposes a polished dashboard at `http://localhost:8501`.

Getting Access

Admin Creates Your Account

Your admin will create your account and provide:

- **Username** and temporary **password**
- Your assigned **namespace** (e.g. `reci-transport`, `ivycandy-hair`)
- Your **role**: client or viewer

You will be prompted to change your password on first login.

Self-Registration (if enabled)

If self-registration is enabled by the admin:

1. Open the dashboard at `http://localhost:8501`
2. Click the **Register** tab on the login screen
3. Enter username, email, password, and select your namespace
4. Log in separately after registration

Password requirements: minimum 10 characters, at least one uppercase letter, one lowercase letter, and one digit.

Roles & Permissions

Feature	Admin	Operator	Client	Viewer	Staff
All namespaces	■	■	Own only	Own only	—
Memory read	■	■	■	■	—
Memory write	■	■	■	■	—
Run agents	■	■	■	■	—
View outputs	■	■	■	■	—
Manage workflows	■	■	■	■	—
View/work tickets	■	■	■	■	Assigned only
Assign & advance tickets	■	■	■	■	Self-assign ■
Admin Panel	■	■	■	■	■
Observability Dashboard	■	■	■	■	■

Staff is a dedicated support role. Staff users see only tickets assigned to them, can self-assign open tickets, and can advance ticket status. They do not have access to memory, agents, outputs, or admin functions.

Dashboard Pages

Overview

The home page shows:

- **System Status** — API and database health at a glance
- **KPI Cards** — memory entries, agents, runs, workflows, outputs
- **Recent Events** — last 8 agent run results with timestamps and eval score pills
- **Quick Dispatch** — run any agent against any namespace with a prompt
- **Memory by Namespace** — count of active memory entries per namespace
- **Live auto-refresh toggle** — when enabled, the feed refreshes every 8 seconds
- **Error alert strip** — red banner displayed when any recent run failed
- **Running-now indicator** — yellow banner showing agents currently executing

Eval score pills appear on each run entry, color-coded:

- Green — score ≥ 4.0
- Amber — score ≥ 2.5
- Red — score < 2.5

Memory

Browse, search, write, and delete memory entries.

- Full-text search across keys and values
- Filter by namespace, category, and minimum confidence
- Write entries with category (fact, preference, reminder, task), TTL, and confidence score
- Reminder entries use a date/time picker for expiry
- Search via FTS5 for instant keyword matching
- Hybrid BM25+vector search available via the search toggle for higher-recall results
- Bulk delete: select entries with checkboxes, then use the bulk toolbar to delete all selected

Agents

The Agents page has three tabs:

Chat tab

Conversational multi-turn chat with any agent. Each session maintains full conversation history per agent. Features:

- **Streaming responses** — tokens appear as they arrive, no waiting for the full output. Fully implemented via Server-Sent Events (SSE) with token-by-token rendering.
- **Image/screenshot upload** — attach images for visual analysis (e.g. UI screenshots, charts, documents)
- **Voice input** — click the microphone button, record audio, and the system auto-transcribes it into the prompt field
- **Conversation history** — full back-and-forth exchange shown per agent; history persists for the session
- **Clear conversation** — button resets the current conversation and starts a new session
- **Eval score badges** — every assistant response shows its quality score after the async eval completes (~10 seconds)
- **Token usage** — token count displayed after each response

Every streaming chat run is saved to run history automatically, with output, tokens, and eval score recorded.

Catalog tab

The original agent grid: all 12 registered agents with their descriptions, capabilities, and a quick-run form. Run any agent with a custom prompt and namespace. Live output polling auto-refreshes every 3 seconds until the run completes.

Run History tab

Full run history across all agents. Now includes:

- Quality score column with color-coded pill
- Eval dimension breakdown (task completion, factual grounding, conciseness, safety) expandable per row
- Per-agent averages in the summary header
- Bulk delete run history entries via checkbox selection

Workflows

- View all 7 configured pipelines
- Trigger workflows manually
- See last-run timestamps and schedules
- Monitor pipeline steps and their statuses
- **Webhooks tab** — enable a webhook URL per workflow so external systems can trigger it:

1. Open a workflow and click the **Webhooks** tab

2. Click **Enable Webhook** — a unique URL and secret are generated
3. Copy the webhook URL and secret
4. POST to the URL from any external system with `X-Webhook-Secret: <secret>` to fire the workflow
5. The webhook endpoint requires no JWT — authentication is via the secret header only

Client Vault

- **Namespaces** — view your namespace details, workspace stats (files, KB), and project count
- **Projects** — list projects in your namespace, update status (active/paused/archived)
- **Context Files** — upload per-namespace context files injected into agent system prompts

Outputs

- Browse all AI-generated outputs: reports, summaries, plans, analyses, code
- Full-text search across output content
- Filter by namespace, type, and date
- Export individual outputs as Markdown or PDF
- Bulk delete: select outputs with checkboxes and delete in one action

Tickets

The ticketing system provides a full support and task lifecycle within ClaudeOS.

- **Create tickets** — title, description, priority (P1–P4), and namespace
- **Priority / SLA tiers:**
 - P1 — Critical (immediate response required)
 - P2 — High
 - P3 — Medium
 - P4 — Low
- **Status flow** — Open → In Progress → Resolved → Closed
- **Assignment** — admin and operator can assign tickets to staff or other users; staff can self-assign open tickets
- **Comments** — add threaded comments to any ticket; comments are loaded on demand when you open the comment panel
- **Resolution notes** — record resolution details when closing a ticket
- **Bulk delete** — select multiple tickets and delete in one action

- **Stats panel** — ticket counts broken down by status and priority

Observability (admin/operator only)

Four sub-tabs providing full system visibility:

Quality Scores

- Per-agent average eval scores displayed as a ranked table
- Distribution chart showing score spread across all runs
- Dimension breakdown: task completion, factual grounding, conciseness, safety
- Low-quality alert: red warning when any agent's average drops below 2.5

Latency

- p50, p95, and p99 latency across all runs
- Per-agent average latency table sorted by slowest
- Time-series chart showing latency trends over the past 7 days

Token Cost

- Total input and output token counts across all runs
- Estimated USD cost based on current claude-sonnet-4-6 pricing
- Per-agent breakdown: tokens used, cost, run count, avg cost per run

Memory Health

- Entry counts per namespace: total, active, consolidated, expired
- Storage size estimate per namespace
- Manual consolidation trigger button — runs the consolidation job immediately rather than waiting for the scheduled 4-hour window
- Visual indicator when any namespace has not been consolidated in over 24 hours

Settings (Admin/Operator)

- System configuration
- Supabase sync controls — trigger manual push or view sync status
- View environment and runtime info

Admin Panel (Admin only)

- **Users** — create, deactivate, unlock, and reset passwords for all users
- **API Keys** — create and revoke API keys; raw key shown once on creation

- **Audit Log** — paginated event log: logins, failures, lockouts, user actions
- **Sessions** — view active refresh token sessions, revoke any session
- **Security** — configure lockout threshold, lockout duration, token TTLs, enable/disable self-registration

Theme

A dark/light mode toggle sits in the **bottom-left corner** of every page, including the login screen. It is a circular icon button (44px) with no text label:

- **Dark mode:** dark navy circle with a white sun icon
- **Light mode:** light gray circle with a dark crescent moon icon

Your theme preference persists for your current session.

Security Features

- **JWT-based authentication** — tokens expire after 60 minutes, auto-refreshed silently
- **Account lockout** — 5 failed login attempts triggers a 15-minute lockout (admin-configurable)
- **Case-insensitive login** — usernames are matched case-insensitively (`Romanus`, `ROMANUS`, and `romanus` all work)
- **Audit log** — every login, logout, failure, user action, and password change is logged with IP and timestamp
- **Namespace isolation** — clients can only read and write within their assigned namespace
- **Bcrypt password hashing** — 12 rounds, never stored in plaintext
- **Forced password change** — admin can flag accounts requiring a new password on next login

Quality Scoring

Every agent run is automatically scored by Claude Haiku (LLM-as-Judge) on 4 dimensions. Scoring is fully asynchronous — it runs in the background after the run completes and does not add latency to the agent response.

Dimension	Scale	Weight	What it measures
Task Completion	0–5	40%	Did the output address the prompt fully?
Factual Grounding	0–5	30%	Are claims grounded in injected memory context?
Conciseness	0–5	20%	Appropriate length, no padding or repetition?
Safety	pass/fail	10%	No harmful, biased, or dangerous content?

Overall score = weighted average of the four dimensions. A safety fail caps the overall score at 1.0 regardless of other scores.

Scores appear:

- In **Run History** as a colored pill next to each run
- In **Overview** as a pill on each recent event entry
- In **Chat** as a badge below each assistant response (appears ~10 seconds after the response completes)
- In **Observability** → **Quality Scores** as aggregated analytics

Low-quality threshold: runs scoring below 2.5 are flagged. The Observability dashboard alerts when any agent's rolling average drops below this threshold.

Advanced Memory

Hybrid BM25 + Vector Search

Memory search uses two retrieval methods simultaneously:

- **BM25** — keyword-based scoring; excellent for exact terms and technical queries
- **Vector (ChromaDB)** — semantic similarity; excellent for concept-level retrieval

Results from both are fused using Reciprocal Rank Fusion (RRF) before being returned. Enable hybrid search via the toggle on the Memory page for higher-recall results.

Memory Consolidation

An automated job runs every 4 hours:

1. Groups semantically similar memory entries per namespace
2. Calls Claude to synthesize each cluster into a single, more concise summary entry
3. Marks the originals as consolidated (they are archived, not deleted)

This keeps memory stores lean and coherent over time. You can also trigger consolidation immediately from **Observability** → **Memory Health**.

Tiered Context Injection

When an agent runs, only the most relevant memory entries are injected into its system prompt, ranked by relevance to the current query. This reduces token consumption by ~40% while maintaining context quality.

MCP Integration (Developers)

ClaudeOS exposes all 12 agents as MCP (Model Context Protocol) tools. Any MCP-compatible client — Claude Desktop, Cursor, custom agents — can call your ClaudeOS agents as first-class tools without writing any custom integration code.

Start the MCP server:

```
.\scripts\start_mcp.ps1
```

The server starts on port 5100. Verify it is running:

```
curl http://localhost:5100/mcp
```

Add to Claude Desktop's config (%APPDATA%\Claude\claude_desktop_config.json):

```
{ "mcpServers": { "claudeos": { "url": "http://localhost:5100/mcp" } } }
```

Restart Claude Desktop. All 12 agents will appear as available tools in the tool picker.

Install optional dependencies first if not already present:

```
pip install mcp uvicorn
```

A2A Agent Cards

Each agent also exposes a machine-readable capability card for agent-to-agent discovery:

```
GET http://localhost:5000/api/v1/agents/<name>/.well-known/agent.json
```

The card describes the agent's name, description, accepted input schema, output schema, and streaming/multi-turn capabilities. External orchestrators can use this endpoint to automatically discover and delegate to ClaudeOS agents.

Webhook API (Developers)

Trigger workflows from external systems without a JWT token.

Step 1 — Enable the webhook

Open the Workflows page, select a workflow, click the **Webhooks** tab, and click **Enable Webhook**. A unique URL and HMAC secret are generated and displayed. Copy both.

Step 2 — Fire the workflow

```
curl -X POST http://localhost:5000/api/v1/workflows/morning-briefing/trigger \
-H "X-Webhook-Secret: your-secret" \
-H "Content-Type: application/json" \
-d '{"context": {"namespace": "global", "topic": "AI news"}}'
```

The `context` object is passed as input to the first pipeline step. The webhook endpoint is public (no JWT required) — authentication is via the `X-Webhook-Secret` header only.

Step 3 — Verify execution

The response returns the run ID. Poll `GET /api/v1/agents/runs/<id>` to check status, or view the run in the **Run History** tab.

SSE Streaming API (Developers)

Stream agent responses token-by-token via Server-Sent Events:

```
curl -N
"http://localhost:5000/api/v1/agents/writing-agent/stream?prompt=Hello&namespace=global" \
-H "X-API-Key: your_api_key"
```

Event types:

Event type	Payload	Description
token	<code>{"type": "token", "text": "..."} </code>	One text chunk as it arrives
done	<code>{"type": "done", "run_id": "...", "tokens_in": N, "tokens_out": N}</code>	Stream complete; includes run ID and token counts
error	<code>{"type": "error", "message": "..."} </code>	An error occurred; run marked as failed

Every stream run is saved to the database automatically — the run appears in Run History with `status=done`, output, and token counts recorded.

API Access (Developers)

The REST API runs at `http://localhost:5000/api/v1/`.

Authenticate with JWT:

```
# Login curl -X POST http://localhost:5000/api/v1/auth/login \ -H "Content-Type: application/json" \ -d '{"username":"your_user","password":"your_pass"}' # Use token curl http://localhost:5000/api/v1/agents \ -H "Authorization: Bearer <access_token>"
```

Or use an API key (scripts/automation):

```
curl http://localhost:5000/api/v1/memory \ -H "X-API-Key: your_api_key"
```

Key Endpoints

Method	Endpoint	Description
POST	<code>/auth/login</code>	Login, get access + refresh tokens
POST	<code>/auth/refresh</code>	Renew access token using refresh token

POST	<code>/auth/logout</code>	Revoke current session
GET	<code>/auth/me</code>	Current user info
GET	<code>/memory</code>	List memory entries
POST	<code>/memory</code>	Write a memory entry
POST	<code>/memory/consolidate</code>	Trigger memory consolidation immediately
GET	<code>/memory/hybrid-search</code>	Hybrid BM25+vector search with RRF reranking
GET	<code>/agents</code>	List all agents
POST	<code>/agents/{name}/run</code>	Dispatch agent run (async, returns run_id)
GET	<code>/agents/runs/{id}</code>	Poll run status and output
GET	<code>/agents/runs</code>	List all runs (filterable by namespace/status)
GET	<code>/agents/{name}/stream</code>	SSE streaming response (token-by-token)
GET	<code>/agents/{name}/.well-known/agent-alias</code>	Agent Card
GET	<code>/agents/{name}/conversations</code>	List multi-turn conversation sessions
POST	<code>/agents/{name}/runs/{id}/cancel</code>	Cancel a pending/running run
GET	<code>/outputs</code>	List outputs
DELETE	<code>/outputs/bulk</code>	Bulk delete outputs
GET	<code>/tickets</code>	List tickets
POST	<code>/tickets</code>	Create a ticket

GET	<code>/tickets/{id}</code>	Get ticket detail
PUT	<code>/tickets/{id}</code>	Update ticket (status, assignee, resolution)
DELETE	<code>/tickets/{id}</code>	Delete a ticket
DELETE	<code>/tickets/bulk</code>	Bulk delete tickets
GET	<code>/tickets/stats</code>	Ticket counts by status and priority
GET	<code>/workflows</code>	List workflows
POST	<code>/workflows/{name}/run</code>	Trigger workflow manually
POST	<code>/workflows/{name}/trigger</code>	Webhook trigger (public, X-Webhook-Secret auth)
POST	<code>/workflows/{name}/webhook/enable</code>	Enable webhook + generate secret
GET	<code>/system/status</code>	System health detail (auth required)
GET	<code>/health</code>	Public health check (no auth)

Starting the System

```
.\scripts\start.ps1
```

This kills any existing processes on :5000 and :8501, starts the Flask API via waitress, verifies `/health`, then starts the Streamlit dashboard.

Start MCP server (optional):

```
.\scripts\start_mcp.ps1
```

First-time setup:

```
pip install -r requirements.txt python scripts/migrate.py python scripts/seed_agents.py python scripts/seed_workflows.py python scripts/seed_namespaces.py python scripts/create_admin.py --username admin --password Admin123! .\scripts\start.ps1
```

Cloud Sync (Supabase)

Outputs and memory can be synced to Supabase for cloud backup and sharing.

1. Create a Supabase project
2. Run `sync/supabase_schema.sql` in the Supabase SQL Editor
3. Add credentials to `.env`:

...

`SUPABASE_URL=https://your-project.supabase.co`

`SUPABASE_SERVICE_KEY=your_service_role_key`

...

1. Restart the server — auto-sync fires every 15 minutes
2. Or trigger manually from the **Settings** tab

Troubleshooting

Problem	Solution
Can't reach dashboard	Run <code>.\scripts\start.ps1</code> — check both ports are alive
Login fails	Check username/password (login is case-insensitive); after 5 failures wait 15 min or ask admin to unlock
"Must change password" shown	Enter current password + new password in the prompt
API returns 401	Token expired — log out and log back in

Output won't export	Try Markdown export; PDF may need wkhtmltopdf installed
Sync fails	Verify SUPABASE_URL and SUPABASE_SERVICE_KEY in .env; restart after changes
Ticket comments not loading	Click the "■ Comments" toggle — comments load on demand
Eval scores not appearing	Scores are async — wait ~10s after run completes; check API logs for Haiku errors
Streaming returns error	Ensure only one Flask process is on port 5000; restart with <code>.\scripts\start.ps1</code>
Voice input not working	Run <code>pip install openai-whisper</code> ; first use downloads ~140MB model automatically
MCP server not found	Run <code>pip install mcp uvicorn</code> then <code>.\scripts\start_mcp.ps1</code>
Runs stuck as "running"	Restart server; stuck runs from crashed sessions are auto-cleaned on next health check
Agents not responding	Check ANTHROPIC_API_KEY in .env; verify it is not expired or rate-limited
ChromaDB slow on first start	Normal — sentence-transformers model loads on first request (~30-60s cold start)

Support

Contact your admin (Romanus Igwe) or open an issue in the project repository.

Stack versions: Python 3.11+ · Flask 3.0 · Streamlit 1.38 · SQLite FTS5 · ChromaDB · claude-sonnet-4-6 · Claude Haiku (eval) · rank-bm25 · plotly · waitress (WSGI)